

# Lecture 10: semiconductors

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MAT SCI 120 — Winter 2026

## Course recap so far

- ▶ Drude (particle) model explains conductivity in metals, but fails for insulators.
- ▶ Particle model fails in other things; solved by the wave–particle duality and Schrödinger equation
- ▶ First, we saw free electrons, confinement, hydrogen.
- ▶ Then, we modeled crystals as periodic potentials (Kronig–Penney) + Bloch theorem → band structures + gaps.
- ▶ Band gap intuition: math, diffraction (Bragg planes), standing waves.
- ▶ Transport:  $E(k)$  → effective mass (mobility); band filling → carriers and conductivity.
- ▶ Today: use the past lectures to explain semiconductors.

## Remembering: conductivity of materials from Drude's model

**Ohm's law:**  $J = \sigma E$  identifies:

$$\sigma = \frac{N_e e^2 \tau}{m}$$

*Notation:*  $N_e$  = electron density,  $\sigma$  = electrical conductivity,  $N_e$  = concentration of electrons,  $\tau$  = mean time-to-collision,  $m$  = electron mass,  $e$  = electron charge

### Problems:

- ▶ We cannot explain semiconductors or insulators with this model
- ▶ We cannot explain mobility and  $N_e$  very well in quantum mechanics
- ▶ How to really explain where electrical properties of materials come from?

### Goals

First, let's find out how electrons move in materials according to the physics of waves.  
Then, let's redefine the dependence on  $N_e$ .

# Wave packets in lattices explain particle-like dynamics of electrons

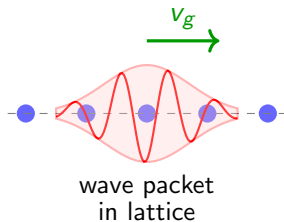
- ▶ Electrons in crystals are **wave packets** with group velocity

$$v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k}$$

- ▶ An external electric field creates a force  $F$  on electrons, changing their momentum:

$$\hbar \frac{dk}{dt} = F$$

- ▶ Last lecture: connected (quantum mechanical) band structure to classical  $F = ma$  dynamics



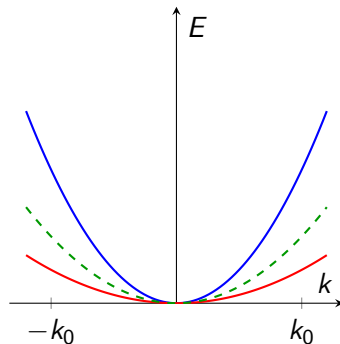
# Effective mass depends on band curvature

## Effective mass:

$$m^* = \hbar^2 \left( \frac{\partial^2 E}{\partial k^2} \right)^{-1}$$

- ▶ **Large curvature**  $\rightarrow$  small  $m^*$   $\rightarrow$  electrons accelerate easily
- ▶ **Small curvature** (flat band)  $\rightarrow$  large  $m^*$   $\rightarrow$  sluggish response
- ▶ Free electron:  $E = \frac{\hbar^2 k^2}{2m}$  gives  $m^* = m$  exactly

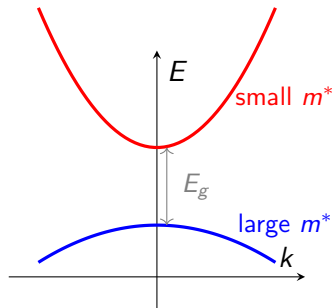
— Small  $m^*$     — Large  $m^*$     - - - Free  $e^-$



Steeper curvature = lighter effective mass = faster response

# Effective mass simplifies the problem of conductivity in materials

- ▶ Near band extrema,  $E(k)$  is **parabolic**
- ▶ Electrons behave like “free particles” with effective mass  $m^*$
- ▶ Crystal potential effects  $\rightarrow$  single parameter
- ▶ Newton's equations are valid:  $F = m^* a$
- ▶  $m^*/m_e$  can be very large or very small!



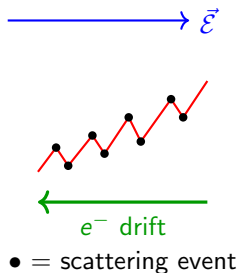
# Remembering the drift velocity from the Drude model

**Notation:**  $v_D$  = drift velocity,  $\mu$  = mobility,  $\tau$  = collision time,  $m^*$  = effective mass

- ▶ Applied field  $\mathcal{E}$  accelerates carriers between scattering events
- ▶  $v_D$  is an “average velocity” between scattering events
- ▶  $v_D \propto \mathcal{E}$  at low fields
- ▶ Typical:  $v_{th} \sim 10^5$  m/s in metals,  $v_D \ll v_{th}$

## Drift velocity

$$v_D = \frac{e\tau}{m^*} \mathcal{E} \quad (1)$$



## Redefining mobility using the effective mass $m^*$

**Notation:**  $\mu$  = mobility [ $\text{m}^2/(\text{V}\cdot\text{s})$ ],  $e$  = elementary charge,  $\tau$  = mean collision time

- ▶ Mobility  $\mu = v_D/\mathcal{E}$  measures how easily carriers move under field
- ▶ Units:  $\text{m}^2/(\text{V}\cdot\text{s})$ , like diffusion coefficient; higher  $\mu \rightarrow$  faster movement of charge carriers
- ▶ Depends on effective mass  $m^*$  and mean collision time  $\tau$

### Mobility as a function of effective mass

$$\mu = \frac{e\tau}{m^*} \quad (2)$$

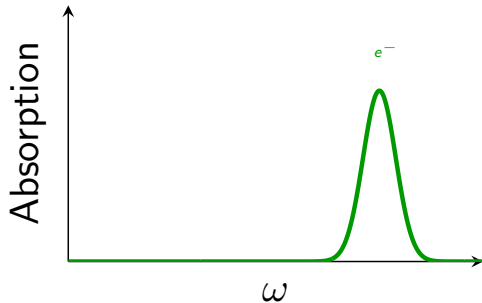


# Measuring effective mass

- ▶ Carriers in field  $B$  orbit at cyclotron frequency:

$$\omega_c = \frac{eB}{m^*} \Rightarrow m^* = \frac{eB_{\text{res}}}{\omega_c}$$

- ▶ Microwave absorption peaks at resonance:  
 $\omega_{\mu\text{w}} = \omega_c$
- ▶ Requires low  $T$ , high purity:  $\omega_c \tau \gg 1$
- ▶ Multiple peaks  $\rightarrow$  distinct  $m^*$  values



## Remembering: conductivity of materials from Drude's model

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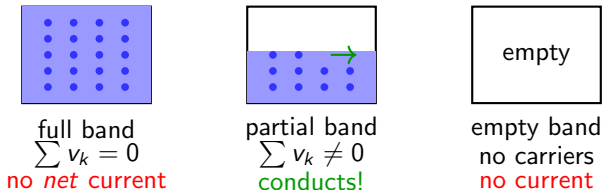
### Resolving issues so far:

- ▶ Different elements have different structures and potentials  $\rightarrow$  different bands ✓
- ▶ We can explain mobility from band structures ✓
- ▶ How to explain  $N_e$  in quantum mechanics?
- ▶ How to really explain where electrical properties of materials come from?

### Goals

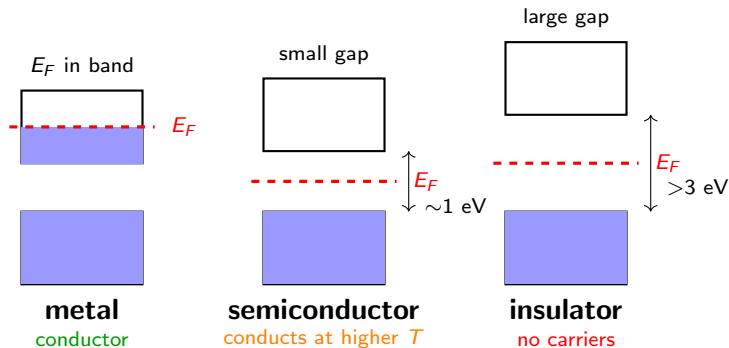
We saw how electrons move in materials (effective mass). Now, let's explain where  $N_e$  comes from.

## Band filling determines conductivity



- ▶ **Full bands:** All  $k$ -states occupied  $\Rightarrow \sum v_k = 0$  (electrons cannot change their directions  $k$  under an electric field)
- ▶ **Partial bands:** Electrons can redistribute their “velocity directions”  $k \Rightarrow$  conduction
- ▶ Fermi level  $E_F$  indicates whether a band is full or not

# Materials and their band diagrams

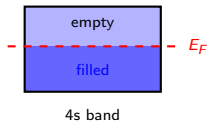


**General rule:** electrical behavior is explained by mobility and band filling  $\rightarrow$  band filling comes from the gap and  $E_F$

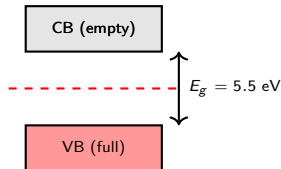
# Metals and insulators: Cu vs diamond

- ▶ **Copper:** Partially filled 4s band,  $E_F$  inside band  $\rightarrow$  excellent conductor
- ▶ **Diamond:**  $E_g = 5.5$  eV,  $k_B T \ll E_g \rightarrow$  negligible excitation, insulator
- ▶ Band theory explains  $\sim 10^{20}$  difference in conductivity!
- ▶ Effective mass in Cu  $\approx m_e$ ; irrelevant in insulators at  $T = 300$  K

**Copper (Metal)**



**Diamond (Insulator)**



## Conceptual questions: band structures

### Test your understanding:

1. **Q1:** Why does a full band contribute zero current even with an applied field?
2. **Q2:** Why does a hole have a negative effective mass?
3. **Q3:** Can a semiconductor conduct at 0 K? Justify this using the Fermi-Dirac distribution
4. **Q4:** Where is the Fermi level in a metal?
5. **Q5:** What if  $|m_h^*| \ll |m_e^*|$ ? What do you expect for the carrier that dominates the conduction of this system?

## Problem we are trying to solve right now

- ▶ We already showed that mobility depends on the curvature of the bands
- ▶ The curvature of the bands depends on the strength/width of the periodic wells in the K-P model
- ▶ Now, we know the conductivity of materials depends on  $\mu$  and  $N_e$
- ▶ We also explained that the Fermi level provides a qualitative sense of  $N_e$  in metals

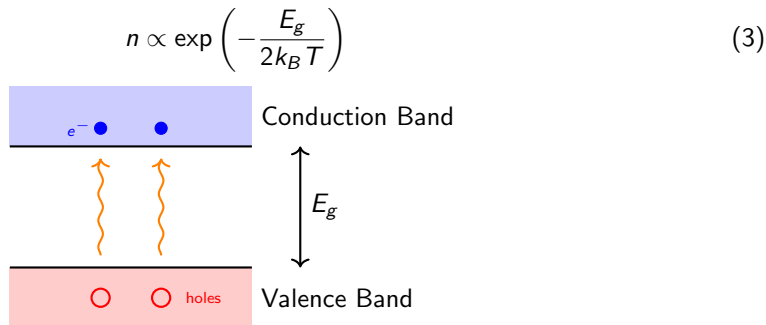
### Main goal

We know that electrons can go to the conduction band and be “free” in semiconductors. How do we compute how many electrons can actually be free?

## Thermal excitation across band gaps

**Notation:**  $E_g$  = band gap;  $k_B$  = Boltzmann constant;  $n$  = carrier concentration

- ▶ At  $T > 0$ , thermal energy  $k_B T$  can excite electrons across band gap  $E_g$
- ▶ Probability of excitation proportional to  $\exp(-E_g/2k_B T)$
- ▶ Creates **electron-hole pairs**: electrons in CB, holes in VB
- ▶ Conductivity increases exponentially with temperature in semiconductors





# Fermi-Dirac statistics and Boltzmann approximation

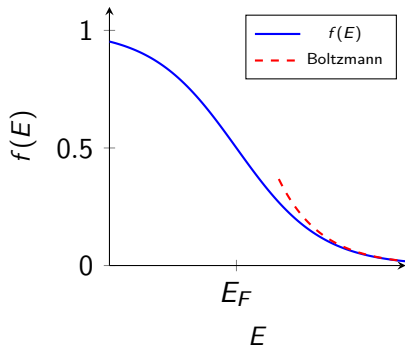
- Occupation probability:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

- When  $E_F$  lies in gap far from edges:

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right)$$

- Valid for lower temperatures  
( $E - E_F \gg k_B T$ )

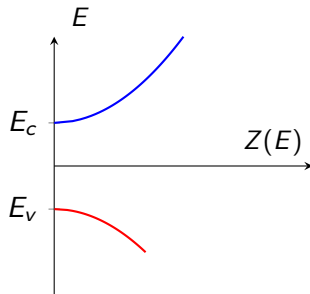


## Remember: density of states in 3D parabolic bands

- ▶ Near band edges,  $E(\mathbf{k})$  is parabolic
- ▶ DOS counts states per unit energy:

$$Z(E) = \frac{1}{2\pi^2} \left( \frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}$$

- ▶ Effective mass  $m^*$  accounts for band curvature



## Deriving electron concentration in conduction band

- ▶ Integrate DOS  $\times$  occupation over conduction band:

$$N_e = \int_{E_c}^{\infty} Z(E) f(E) dE$$

- ▶ Using Boltzmann approximation:

$$N_e = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

- ▶  $N_c$  is the number of carriers in the conduction band, given by:

$$N_c = 2 \left( \frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2}$$

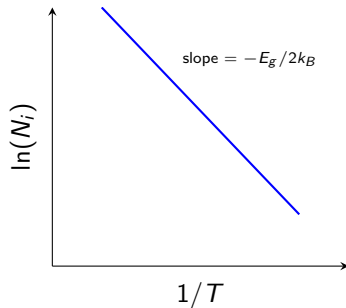
- ▶  $N_c \propto T^{3/2}$  captures thermal broadening of available states

## Intrinsic carrier concentration: Arrhenius dependence with $T$

- ▶ Intrinsic:  $N_e = N_h = N_i$
- ▶ Thermal excitation creates electron-hole pairs

$$N_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$$

- ▶ Exponential dominates over  $T^{3/2}$  prefactor
- ▶ **Unlike metals:** conductivity increases with  $T$



## Conceptual questions: band structures

### Reflect and discuss:

1. **Q1:** Do you expect the concentration of carriers to change with  $T$  in metals?
2. **Q2:** Why do we have to multiply the density of states  $Z(E)$  by the Fermi-Dirac distribution  $f(E)$  to get the carrier concentration?
3. **Q3:** What are the “engineering knobs” that we have available to change the conductivity of a material?

## Problem we are trying to solve right now

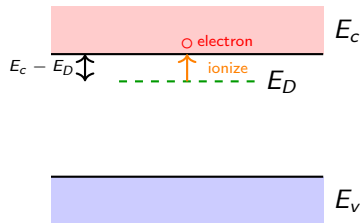
- ▶ We showed that  $N_e$  depends on the band gap and the temperature.
- ▶ In intrinsic semiconductors, the number of electrons in the conduction band  $N_c$  is equal to the number of holes in the valence band  $N_v$ ,  $N_c = N_v = N_i$

### Main goal

Engineering is about controlling nature. How do we control the number of carriers in a semiconductor and, with it, control its conductivity?

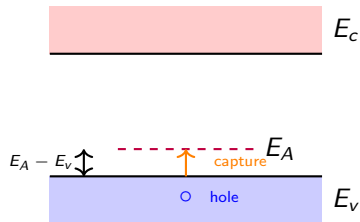
## Donor impurities: adding electrons to the conduction band

- ▶ Group V atoms (P, As, Sb) in Si provide extra valence electron
- ▶ Extra electron loosely bound in hydrogen-like orbit
- ▶ Donor level  $E_D$  lies just below  $E_c$  (typically 10–50 meV)
- ▶ At room  $T$ :  $k_B T \approx 26$  meV ionizes most donors



## Acceptor impurities: adding holes to the valence band

- ▶ Group III atoms (B, Al, Ga) in Si: one fewer valence electron
- ▶ Acceptor captures electron from valence band  $\rightarrow$  mobile hole
- ▶ Acceptor level  $E_A$  lies just above  $E_v$  (10–50 meV)
- ▶ Ionization: neutral atom  $\rightarrow A^- + \text{hole}$





## Donor ionization energy: hydrogen-like atom

- ▶ Donor is a hydrogen-like atom in dielectric medium with permittivity  $\epsilon$  and effective mass  $m^*$
- ▶ Bohr model scaled: binding energy reduced by factor  $(m^*/m_0)(1/\epsilon^2)$

$$E_D = \frac{m_e^*}{m_0} \cdot \frac{1}{\epsilon^2} \cdot 13.6 \text{ eV}$$

- ▶ **Example for Si:**  $\epsilon \approx 12$ ,  $m^* \approx 0.3 m_0$

$$E_D \approx \frac{0.3}{144} \times 13.6 \text{ eV} \approx 30 \text{ meV}$$

- ▶ Compare to hydrogen: 13.6 eV  $\rightarrow$  shallow donors easily ionized

## Charge neutrality condition

- ▶ Crystal must remain electrically neutral
- ▶ **Positive charges:** holes ( $N_h$ ) and ionized donors ( $N_D^+$ )
- ▶ **Negative charges:** electrons ( $N_e$ ) and ionized acceptors ( $N_A^-$ )

$$N_e + N_A^- = N_h + N_D^+$$

- ▶ This constraint determines Fermi level position for any doping profile
- ▶ Combined with mass action law  $N_e N_h = N_i^2$ , enables calculation of all carrier densities

$$\begin{array}{ll} \text{n-type:} & N_D^+ \gg N_A^- \Rightarrow N_e \approx N_D^+ \\ \text{p-type:} & N_A^- \gg N_D^+ \Rightarrow N_h \approx N_A^- \end{array}$$

# Mass action law is a thermodynamic equilibrium constraint

**Notation:**  $N_e$  = electron concentration,  $N_h$  = hole concentration,  $N_i$  = intrinsic carrier concentration

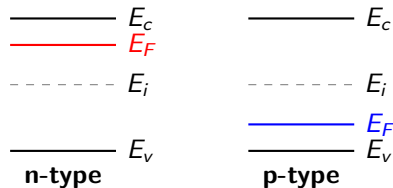
- ▶ Product of electron and hole concentrations fixed at given temperature
- ▶ Holds regardless of doping level—adding donors increases  $N_e$ , decreases  $N_h$
- ▶ Reflects dynamic equilibrium: thermal generation rate = recombination rate
- ▶ Combined with charge neutrality, enables solving for  $N_e$ ,  $N_h$ , and  $E_F$

## Mass Action Law

$$N_e N_h = N_i^2 \quad (4)$$

## Fermi level shift with doping

- ▶ **n-type** ( $N_D \gg N_A$ ):  $E_F$  moves toward  $E_c$ ;  
 $N_e \approx N_D$
- ▶ **p-type** ( $N_A \gg N_D$ ):  $E_F$  moves toward  $E_v$ ;  
 $N_h \approx N_A$
- ▶ Higher doping  $\rightarrow E_F$  closer to band edge
- ▶ Temperature affects ionization fraction



# Summary

- ▶ We explained that electrons and holes have an effective mass, allowing us to simplify our formalism.
- ▶ Effective masses allow us to simplify the problem to a “free electron” moving with an effective mass, thus producing a mobility.
- ▶ We computed  $N_c$ ,  $N_v$  as a function of the temperature, gap, and the dopant concentration.

## Main goal

What additional characteristics should we know to engineer devices?